

Day 92 Project

Simulating Stock Prices using GBM & Applying Ito's Lemma

1 Introduction

In this project, we simulate stock prices using Geometric Brownian Motion (GBM) and apply Ito's Lemma to transform the process.

GBM is widely used in financial modeling, particularly in the Black-Scholes framework.

2 Geometric Brownian Motion

The GBM model is defined as:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

where:

- μ = drift
- σ = volatility
- W_t = Brownian Motion

3 Solution of GBM

The analytical solution is:

$$S_t = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right)$$

4 Applying Ito's Lemma

Let:

$$f(S_t) = \ln S_t$$

Using Ito's Lemma:

$$d(\ln S_t) = \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dW_t$$

5 Interpretation

- Log-prices follow a simpler linear stochastic process
- This transformation simplifies analysis
- Returns become normally distributed under GBM

6 Numerical Simulation

We simulate:

- Brownian Motion paths
- GBM stock price paths
- Log-transformed process

7 Conclusion

This project demonstrates how stochastic calculus connects theory and financial modeling.

Ito's Lemma allows transformation of complex stochastic processes into more tractable forms.